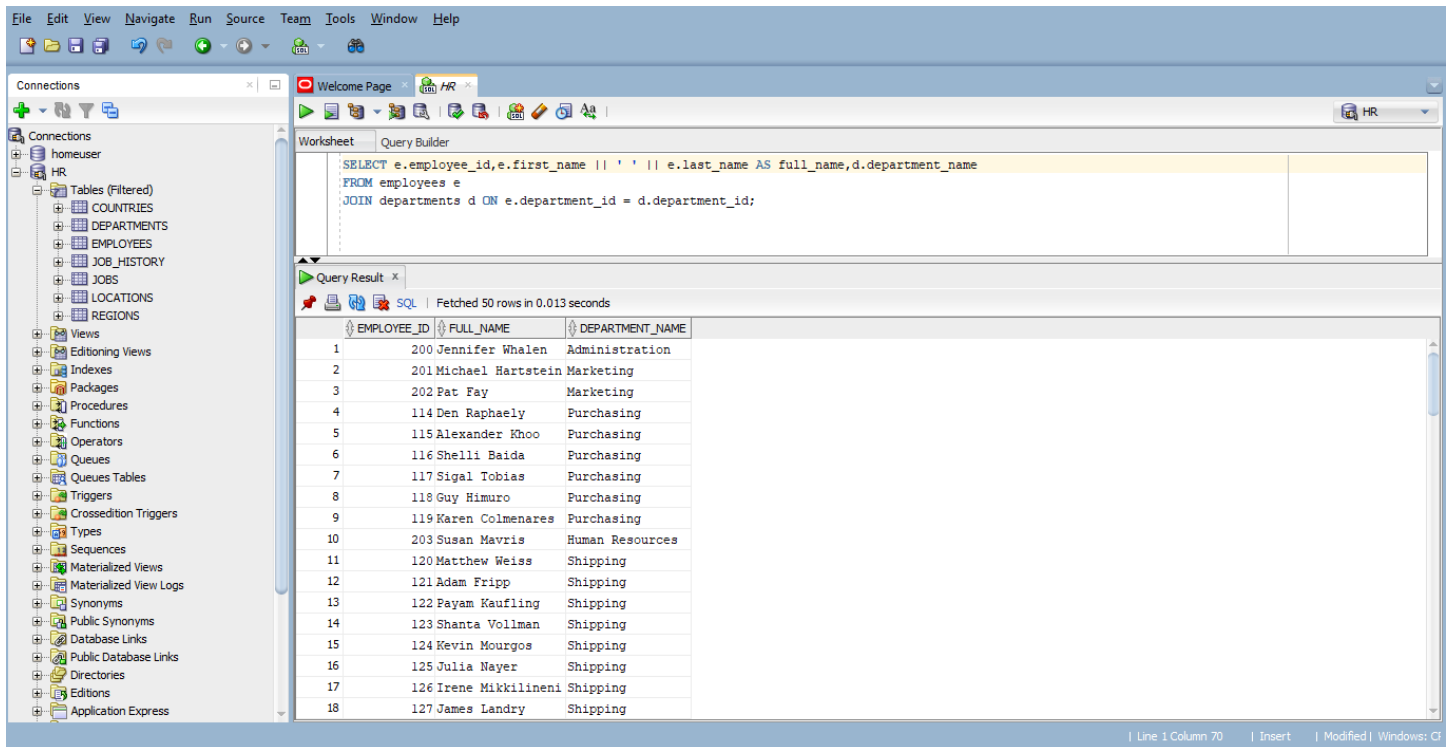


Section 5: Basic SQL

Write a query to list the names of all employees along with their department names. Include the employee's ID, full name, and department name.



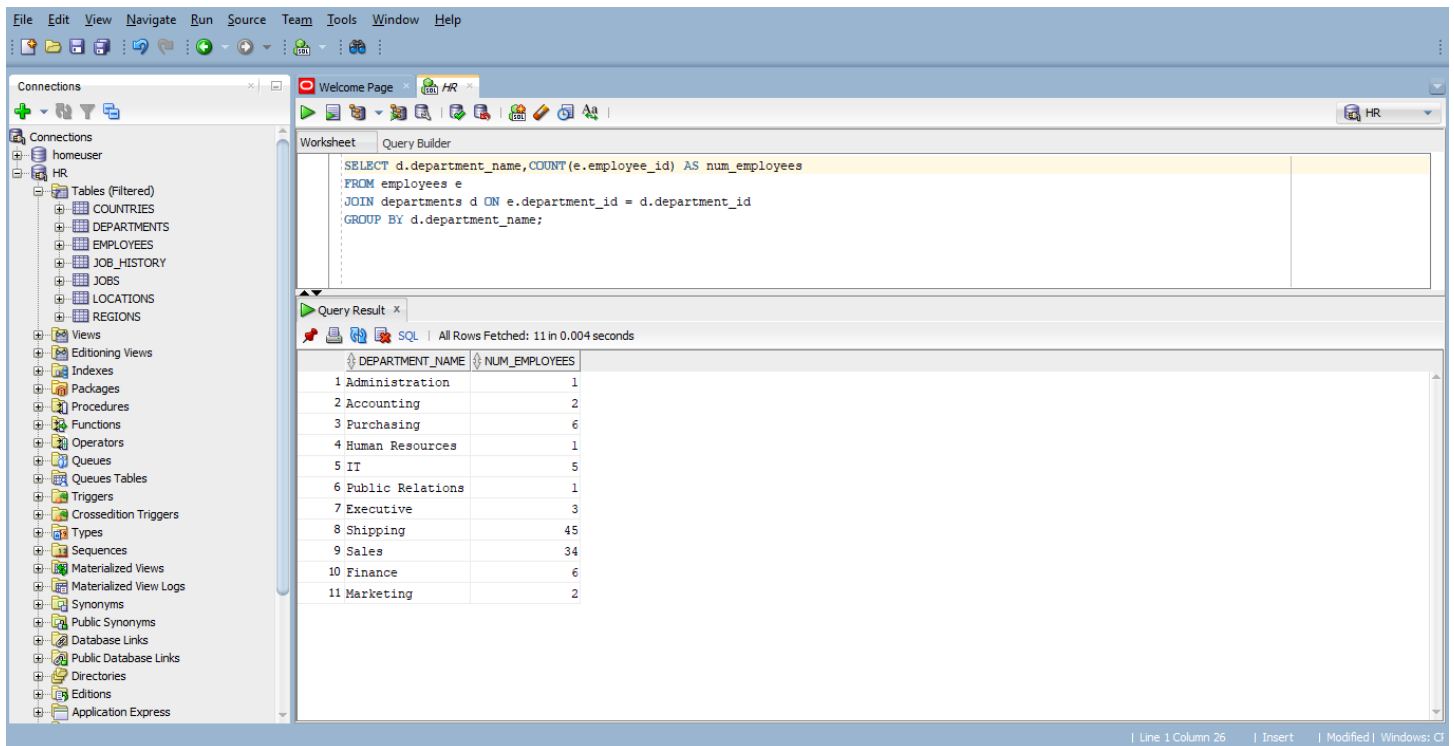
The screenshot shows the SQL Developer interface with a query in the Query Builder window. The query is:

```
SELECT e.employee_id,e.first_name || ' ' || e.last_name AS full_name,d.department_name
FROM employees e
JOIN departments d ON e.department_id = d.department_id;
```

The Query Result window shows 50 rows fetched in 0.013 seconds. The results are as follows:

EMPLOYEE_ID	FULL_NAME	DEPARTMENT_NAME
1	200 Jennifer Whalen	Administration
2	201 Michael Hartstein	Marketing
3	202 Pat Fay	Marketing
4	114 Den Raphaely	Purchasing
5	115 Alexander Khoo	Purchasing
6	116 Shelli Baida	Purchasing
7	117 Sigal Tobias	Purchasing
8	118 Guy Himuro	Purchasing
9	119 Karen Colmenares	Purchasing
10	203 Susan Mavris	Human Resources
11	120 Matthew Weiss	Shipping
12	121 Adam Fripp	Shipping
13	122 Payam Kaufling	Shipping
14	123 Shanta Vollman	Shipping
15	124 Kevin Mourgos	Shipping
16	125 Julia Nayer	Shipping
17	126 Irene Mikkilineni	Shipping
18	127 James Landry	Shipping

Write a query to count the number of employees in each department. Include the department name and the number of employees.



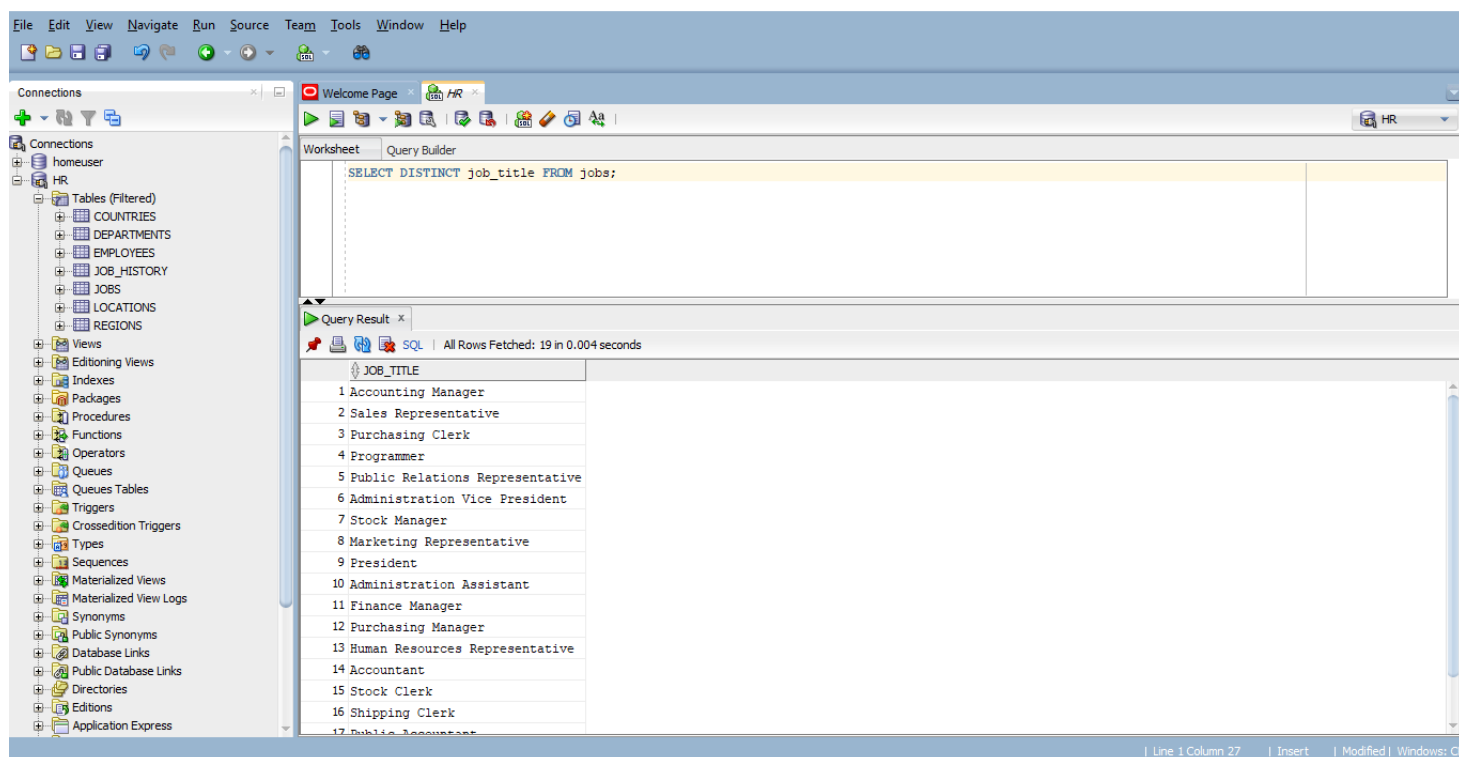
The screenshot shows the SQL Developer interface with a query in the Query Builder window. The query is:

```
SELECT d.department_name,COUNT(e.employee_id) AS num_employees
FROM employees e
JOIN departments d ON e.department_id = d.department_id
GROUP BY d.department_name;
```

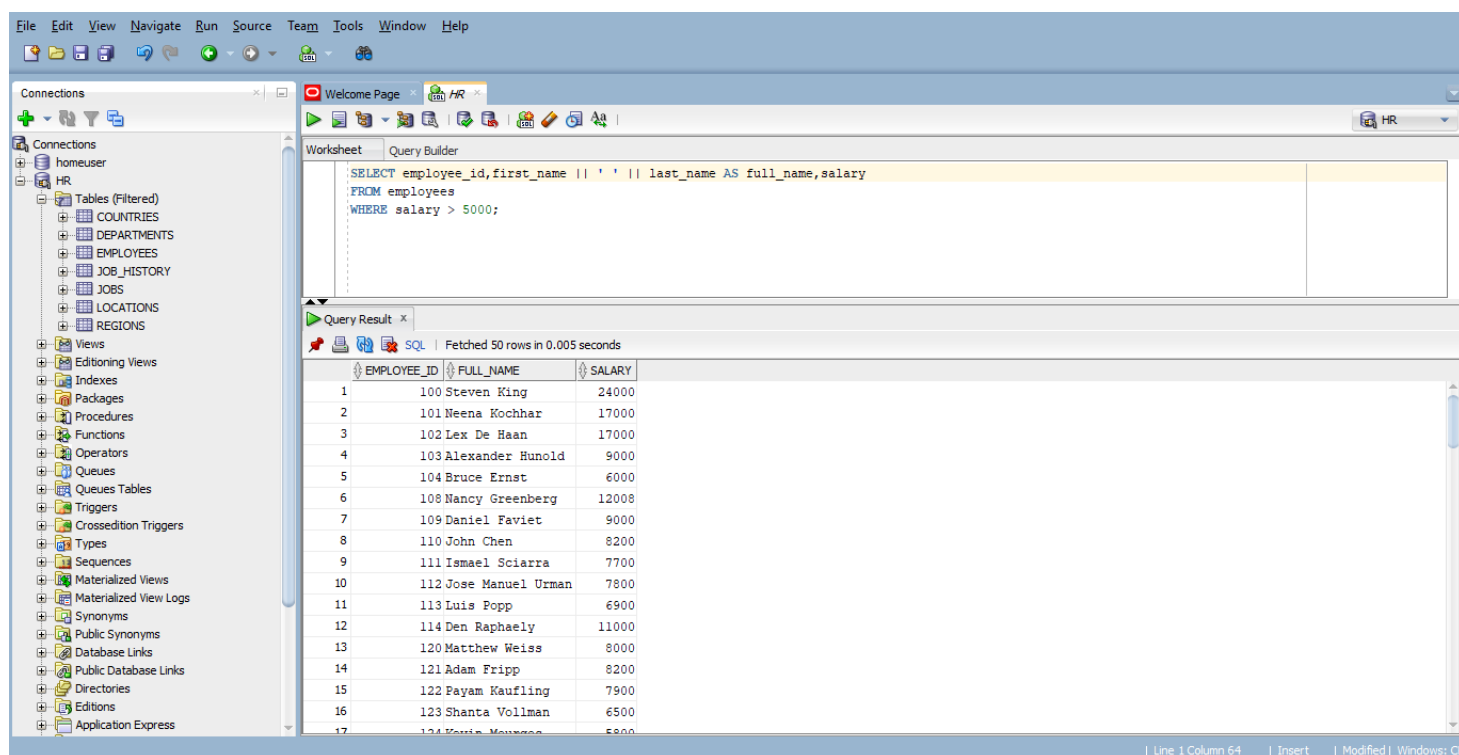
The Query Result window shows all rows fetched in 0.004 seconds. The results are as follows:

DEPARTMENT_NAME	NUM_EMPLOYEES
1 Administration	1
2 Accounting	2
3 Purchasing	6
4 Human Resources	1
5 IT	5
6 Public Relations	1
7 Executive	3
8 Shipping	45
9 Sales	34
10 Finance	6
11 Marketing	2

Write a query to list all job titles available in the company.



Write a query to find all employees who earn more than \$5000 per month. Include the employee's ID, full name, and salary.



Write a query to list the names of all departments along with their location.

The screenshot shows the Oracle SQL Developer interface. On the left, the 'Connections' pane shows a connection to 'homeuser' under the 'HR' schema. The 'Tables (Filtered)' list includes COUNTRIES, DEPARTMENTS, EMPLOYEES, JOB_HISTORY, JOBS, LOCATIONS, and REGIONS. The main window displays a SQL query in the 'Query Builder' tab:

```
SELECT d.department_name, l.city || ', ' || l.state_province AS location
FROM departments d
JOIN locations l ON d.location_id = l.location_id;
```

The 'Query Result' pane shows the results of the query, with 27 rows fetched in 0.005 seconds. The results are displayed in a table with two columns: DEPARTMENT_NAME and LOCATION.

DEPARTMENT_NAME	LOCATION
1 IT	Southlake, Texas
2 Shipping	South San Francisco, California
3 Administration	Seattle, Washington
4 Purchasing	Seattle, Washington
5 Executive	Seattle, Washington
6 Finance	Seattle, Washington
7 Accounting	Seattle, Washington
8 Treasury	Seattle, Washington
9 Corporate Tax	Seattle, Washington
10 Control And Credit	Seattle, Washington
11 Shareholder Services	Seattle, Washington
12 Benefits	Seattle, Washington
13 Manufacturing	Seattle, Washington
14 Construction	Seattle, Washington
15 Contracting	Seattle, Washington
16 Operations	Seattle, Washington
17 IT Support	Seattle, Washington

Write a query to find the earliest hire date in the company. Just show the date.

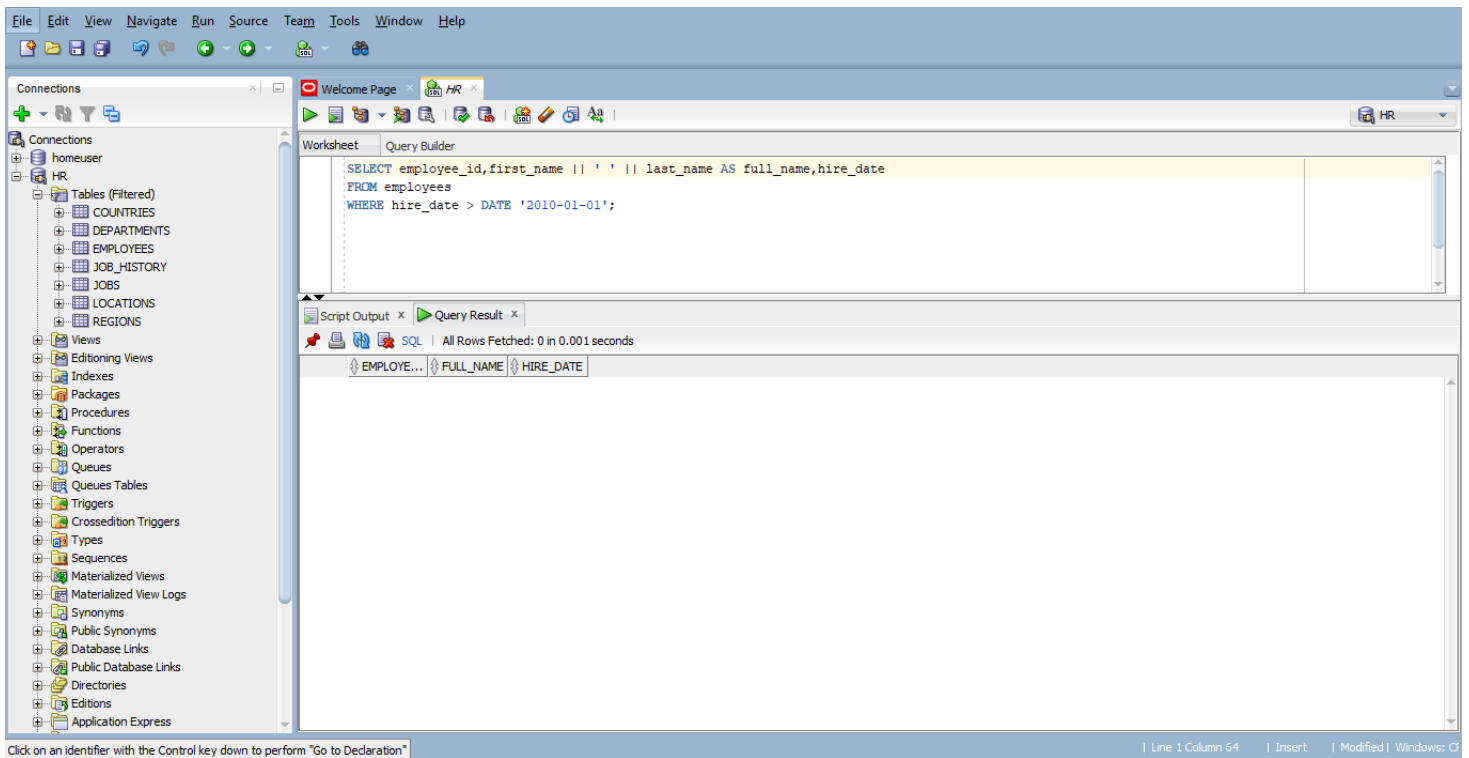
The screenshot shows the Oracle SQL Developer interface. The 'Query Builder' tab displays a SQL query:

```
SELECT MIN(hire_date) AS earliest_hire_date FROM employees;
```

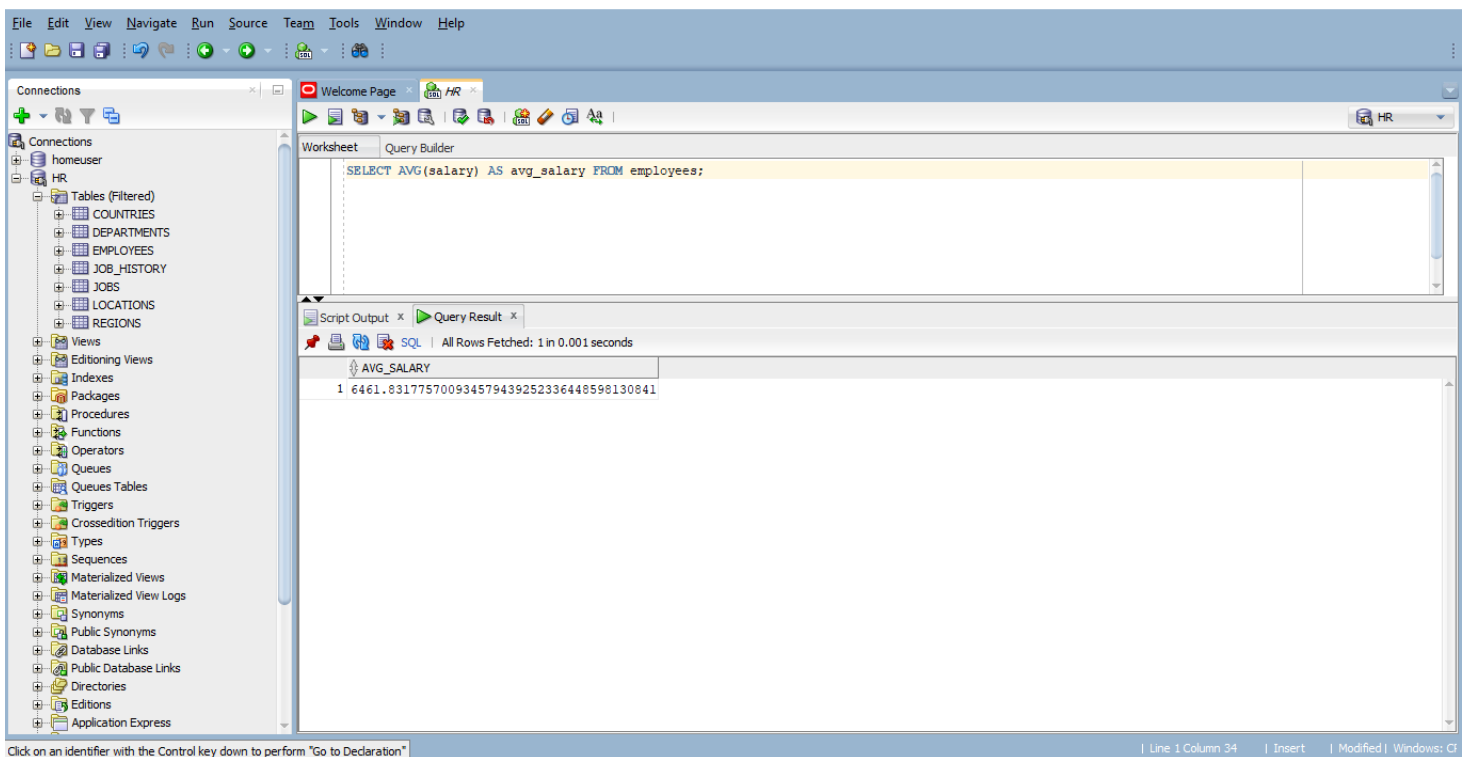
The 'Query Result' pane shows the results of the query, with 1 row fetched in 0.002 seconds. The results are displayed in a table with one column: EARLIEST_HIRE_DATE.

EARLIEST_HIRE_DATE
1 13-JAN-01

Write a query to list all employees who were hired after January 1, 2010. Include the employee's ID, full name, and hire date.



Write a query to calculate the average salary of all employees in the company.



Write a query to list all department names.

The screenshot shows the Oracle SQL Developer interface. The 'Connections' pane on the left lists the 'HR' schema. The 'Worksheet' pane contains the following SQL query:

```
SELECT department_name FROM departments;
```

The 'Query Result' pane displays the results of the query:

DEPARTMENT_NAME
1 Administration
2 Marketing
3 Purchasing
4 Human Resources
5 Shipping
6 IT
7 Public Relations
8 Sales
9 Executive
10 Finance
11 Accounting
12 Treasury
13 Corporate Tax
14 Control And Credit
15 Shareholder Services
16 Benefits
17 Manufacturing

Click on an identifier with the Control key down to perform "Go to Declaration" | Line 1 Column 24 | Insert | Modified | Windows: C

Write a query to find all employees with the job title 'Sales Representative'. Include the employee's ID, full name, and job title.

The screenshot shows the Oracle SQL Developer interface. The 'Connections' pane on the left lists the 'HR' schema. The 'Worksheet' pane contains the following SQL query:

```
SELECT e.employee_id,e.first_name || ' ' || e.last_name AS full_name,j.job_title
FROM employees e
JOIN jobs j ON e.job_id = j.job_id
WHERE j.job_title = 'Sales Representative';
```

The 'Query Result' pane displays the results of the query:

EMPLOYEE_ID	FULL_NAME	JOB_TITLE
1	150 Peter Tucker	Sales Representative
2	151 David Bernstein	Sales Representative
3	152 Peter Hall	Sales Representative
4	153 Christopher Olsen	Sales Representative
5	154 Nanette Cambrault	Sales Representative
6	155 Oliver Tuvault	Sales Representative
7	156 Janette King	Sales Representative
8	157 Patrick Sully	Sales Representative
9	158 Allan McEwen	Sales Representative
10	159 Lindsey Smith	Sales Representative
11	160 Louise Doran	Sales Representative
12	161 Sarath Sewall	Sales Representative
13	162 Clara Vishney	Sales Representative
14	163 Danielle Greene	Sales Representative
15	164 Mattea Marvins	Sales Representative
16	165 David Lee	Sales Representative

Click on an identifier with the Control key down to perform "Go to Declaration" | Line 1 Column 70 | Insert | Modified | Windows: C